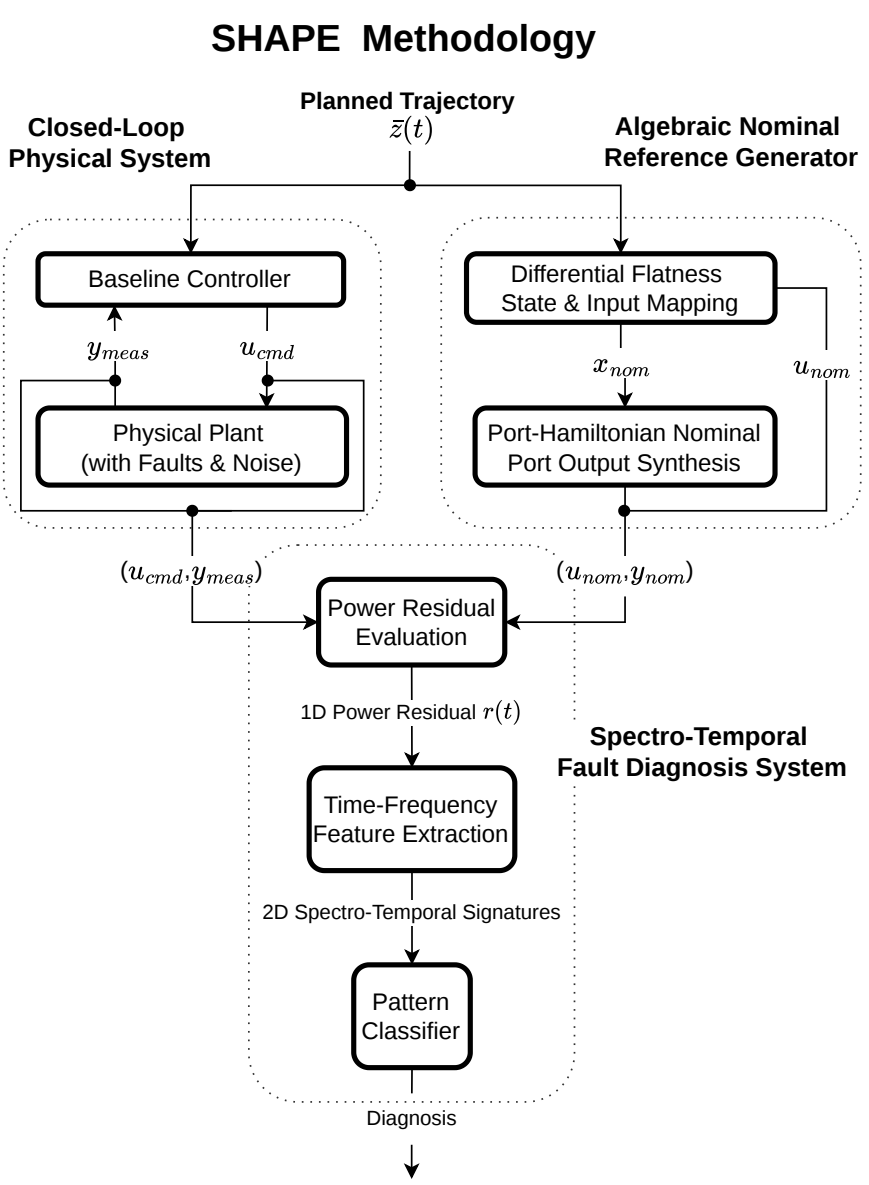

SHAPE Methodology
Planned Trajectory
$\bar{z}(t)$
Closed-Loop
Physical System
Algebraic Nominal
Reference Generator
Baseline Controller
Differential Flatness
State & Input Mapping
y_{meas}
u_{cmd}
x_{nom}
u_{nom}
Physical Plant
(with Faults & Noise)
Port-Hamiltonian Nominal
Port Output Synthesis
(u_{cmd}, y_{meas})
(u_{nom}, y_{nom})
Power Residual
Evaluation
1D Power Residual $r(t)$
Spectro-Temporal
Fault Diagnosis System
Time-Frequency
Feature Extraction
2D Spectro-Temporal Signatures
Pattern
Classifier
Diagnosis